COMPUTATIONAL FLUID DYNAMICS RESEARCH STUDY

Numerical Study of Aerodynamic Wind Forces on Isolated and Arrayed Solar Panels

Steady RANS OpenFOAM (k-ω SST) Parametric Analysis of Wind Velocity, Tilt, Yaw Angles, Multi-Row Sheltering, and Rooftop Integration

CFD Methodology: OpenFOAM simpleFoam | Mesh: snappyHexMesh | Turbulence Closure: k-ω SST

Dataset Root: /home/lavender/OpenFoamUbu/offProjects/solarPanel/solarPanel/

Focus: High-Wind Coastal Resilience & Bluff-Body Aerodynamic Force Quantification | October 2026

Motivation & Failure Modes in Extreme Cyclonic Weather

Recent tropical cyclones in coastal India demonstrate structural vulnerabilities under extreme aerodynamic loads.

The Aerodynamic & Engineering Challenge

- **Solar PV Expansion in High-Wind Coastal Zones**
  India operates over 90 GW of solar PV capacity. Major utility-scale installations are situated across coastal Gujarat, Rajasthan, and Odisha where cyclonic activity is prevalent.
- **Documented Cyclone Damage Rates**
  Field reports from Cyclone Tauktae (May 2021, gusts ~220 km/h), Cyclone Biparjoy (June 2023, ~165 km/h), and Cyclone Fani (May 2019, ~215 km/h) recorded 15% to 20% physical module and mounting failures.
- **Observed Mechanical Damage Modes**
  Structural failure inspections reveal panel frame bending, fastener pulling, purlin detachment, and tracker torque-tube twisting under severe wind loading.
- **Current Code Limitations (IS 875 Part 3)**
  Existing design standards (IS 875 Part 3) prescribe wind pressure for standard enclosed buildings and solid walls, lacking dedicated aerodynamic force coefficients for low-profile tilted PV modules and multi-row wake interferences.



Figure 1a: Cyclone Tauktae Structural Damage (Gujarat, May 2021). Peak wind gusts (~220 km/h) caused catastrophic bending, fastener shear, and torn purlins on ground-mounted tracking arrays.



Figure 1b: Cyclone Biparjoy Module Failures (Coastal Gujarat, June 2023). Extreme turbulent storm loading (~165 km/h) induced torque-tube twisting and widespread panel frame pull-out failures.

Geometric Configuration & Computational Domain Design

Domain dimensions, blockage ratio control, and Volume of Refinement (VOR) specifications.

Domain Sizing & Boundary Conditions

- **Solar Panel Dimensions**
  Length (chord) L = 0.600 m, Width W = 0.500 m, Thickness t = 0.020 m. Mounted at ground clearance height H = 0.500 m.
- **Computational Domain Span (25L × 3W × 3H)**
  Streamwise domain length = 25L (15.0 m total: 5L upstream fetch to prevent inlet blockage, 20L downstream for wake development). Lateral width = 3W (1.8 m, 1W lateral clearance on both sides). Domain height = 3H (1.24 m).
- **Blockage Ratio Minimization (< 3%)**
  Frontal projected area produces a blockage ratio < 3%, well below the 5% threshold recommended by COST 732 / Franke et al. to avoid artificial boundary constriction.
- **Volume of Refinement (VOR)**
  Near-body nested domain: 8.5L length (1.5L upstream, 7.0L wake), 2W width, 2.0 m height to capture steep pressure gradients and recirculating shear layers.
- **Boundary Condition Definitions**
  Inlet: Uniform velocity (3 to 75 m/s), Turbulence Intensity I = 10%.
Outlet: Static gauge pressure p = 0 Pa.
Ground & Panel: No-slip wall boundary.
Ceiling & Sides: Symmetry planes.

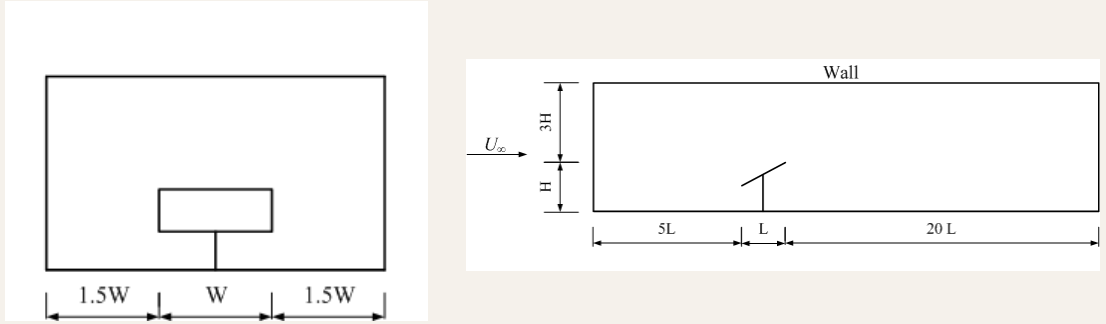


Figure 2: Computational Domain Geometric Sizing. Front cross-section (left) showing 1W lateral side clearances, and longitudinal side view (right) illustrating the 5L upstream fetch and 20L downstream wake recovery zone (blockage < 3%).

Domain Parameter	Geometric Value	Engineering Justification
Streamwise Length	15.00 m (25L)	5L fetch prevents inlet blockage; 20L wake recovery
Lateral Domain Width	1.80 m (3W)	1W clearance prevents lateral constriction on sides
Domain Height	1.24 m (3H)	Maintains blockage ratio < 3% per COST 732 guidelines
Ground Clearance (H)	0.500 m	Replicates standard utility-scale torque-tube mounting
Volume of Refinement	8.5L × 2W × 2.0 m	Refines steep shear gradients and recirculation zone

Mesh Generation & Grid Independence Verification

Asymptotic drag convergence verified across 7 mesh resolutions from 147k to 1.73M cells.

Mesh Level	Cell Count	Drag Fd (N)	Cd	Rel. Change (%)	Status
Mesh 1	147,253	13.238	0.1745	—	Coarse Baseline
Mesh 2	165,341	13.038	0.1718	1.53%	Under-resolved
Mesh 3	183,429	13.018	0.1716	0.15%	Transitional
Mesh 4	**680,590**	**11.282**	**0.1487**	**15.39%**	**Shear Resolved**
Mesh 5	715,391	11.282	0.1487	-0.003%	Asymptotic Plateau
Mesh 6	924,009	11.282	0.1487	+0.001%	Production Standard
Mesh 7	1,735,557	11.282	0.1487	+0.001%	Grid-Independent

Mesh Metrics & Convergence Summary

- **snappyHexMesh Refinement Hierarchy**
  Base background mesh cell size = 60 mm generated with blockMesh. Surface refinement ratio = 0.1 (6 mm); Volume refinement ratio = 0.5 (30 mm) within VOR.
- **Mesh Quality Metrics (Production 912k Mesh)**
  Max Aspect Ratio = 3.46, Max Non-Orthogonality = 36.07° (< 70° limit), Max Skewness = 4.18.
- **Grid Independence Criteria**
  The drag coefficient Cd converges to 0.1487 beyond 680,590 cells. Subsequent variation is less than 0.003%, demonstrating numerical mesh independence.

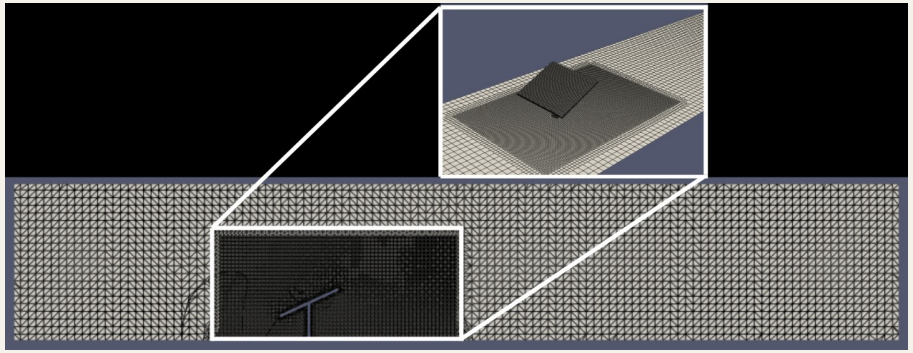


Figure 3a: snappyHexMesh Near-Wall Octree Refinement. Cartesian hexahedral mesh featuring 6 mm surface cells and 30 mm nested volume refinement resolving boundary detachment.

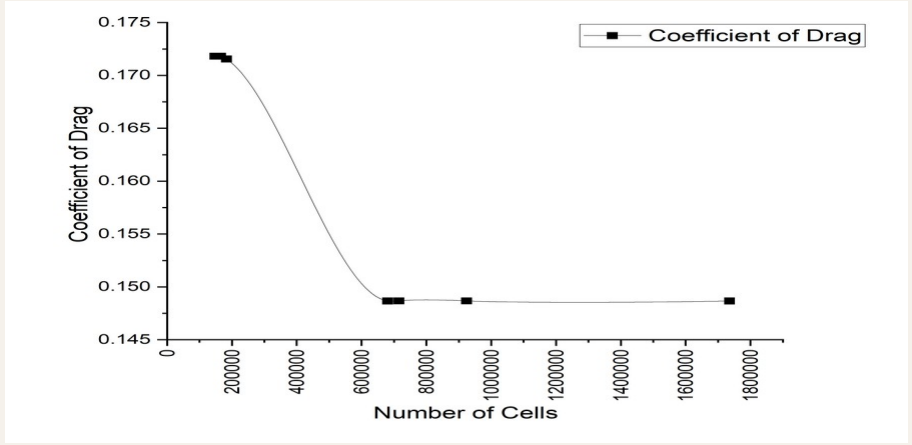


Figure 3b: Grid Convergence Plateau (147k to 1.73M cells). Asymptotic drag stabilization at 680,590 cells (Cd = 0.1487) with variation < 0.003% through 1.73M cells.

Experimental Validation Against Wind Tunnel Benchmark

High-fidelity agreement across 0° to 180° rotation angles with 3.85% mean absolute error.

Yaw Angle ψ	OpenFOAM Cd	Exp. Cd (Yemenici)	Deviation (%)	Observed Aerodynamic Regime
0°	1.0813	1.1973	9.68%	Direct frontal stagnation
30°	0.9412	0.9959	5.50%	Oblique boundary layer detachment
60°	0.5203	0.5388	3.43%	Strong lateral streamline deflection
90°	**0.0055**	**0.0054**	**0.32%**	**Edge-on minimum frontal area**
120°	0.3800	0.3701	2.69%	Onset of leeward recirculation
150°	0.8581	0.8544	0.43%	Rear oblique vortex expansion
180°	1.0867	1.1429	4.91%	Full leeward bluff stagnation

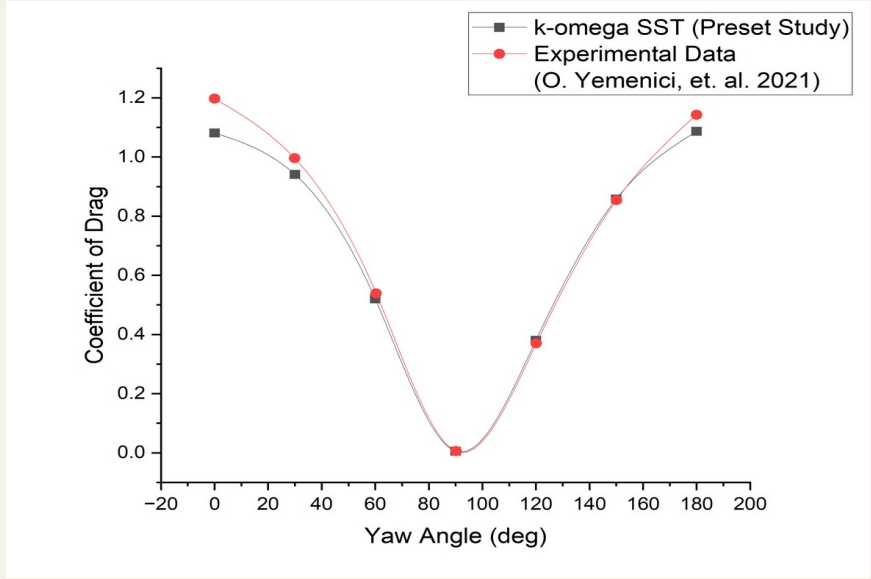


Figure 4: OpenFOAM vs. Yemenici et al. (2021) Validation. Comparison across 7 yaw angles demonstrates 3.85% mean absolute error and 0.32% deviation at 90° edge-on flow.

Validation Findings & Error Analysis

- **High Correlation Across Rotation Angles**
  Mean absolute error across all 7 rotation angles is 3.85%. OpenFOAM accurately reproduces the U-shaped drag profile with symmetric maxima at 0° and 180°.
- **Minimum Drag at 90°**
  At 90° (flow parallel to panel face), Cd drops to 0.0055 in OpenFOAM vs 0.0054 experimentally (0.32% deviation).
- **Error Sources**
  Minor deviation at 0° (~9.7%) is attributed to experimental mounting bracket interference and wind tunnel boundary layers omitted from the isolated computational domain.

3.85%
Mean Absolute Error
Across 7 yaw angles

0.32%
Min Error at 90°
Cd = 0.0055 vs 0.0054

Verified
k-ω SST Closure
Bluff detachment fidelity

Wind Velocity Scaling & Reynolds Number Independence

Forces scale quadratically with velocity (U^2) while aerodynamic drag coefficient plateaus at Re > 50,000.

Reynolds Number Effects & Dynamic Force Scaling

- **Parametric Velocity Sweep (3 to 75 m/s)**
  Simulations span Re = 11,465 (3 m/s) to Re = 286,624 (75 m/s) with characteristic length D = 0.6 m and kinematic viscosity $\nu = 1.57 \times 10^{-5}$ m^2/s.
- **Bluff-Body Re Independence**
  Drag coefficient Cd increases from 0.70 at 3 m/s and stabilizes at ~0.78 beyond Re > 50,000 (15 m/s). Because flow separation is fixed at sharp plate edges, Cd becomes Reynolds-independent at stormy wind speeds.
- **Negative Lift (Downward Force) Dominance**
  At 0° yaw, lift coefficient CL remains consistently negative (CL ≈ -1.22), imposing strong downward compression on the panel face.
- **Dynamic Force Magnitudes ($F \propto U^2$)**
  Drag and lift scale strictly with velocity squared ($F = 1/2\ \rho\ A\ Cd\ U^2$), reaching 340.2 N drag and -650.5 N suction on a single 0.3 m^2 module at 75 m/s.

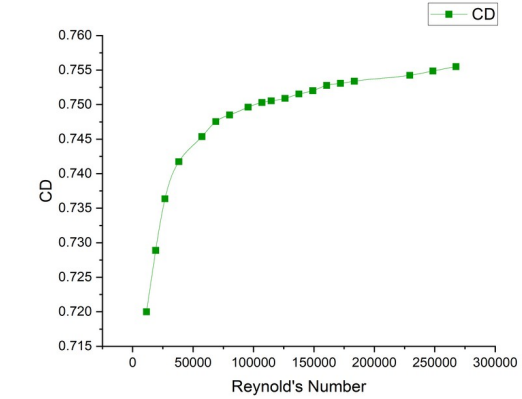


Figure 5a: Cd vs. Reynolds Number. Cd plateaus at ~0.78 for Re > 50,000 due to fixed geometric flow separation at sharp perimeter edges.

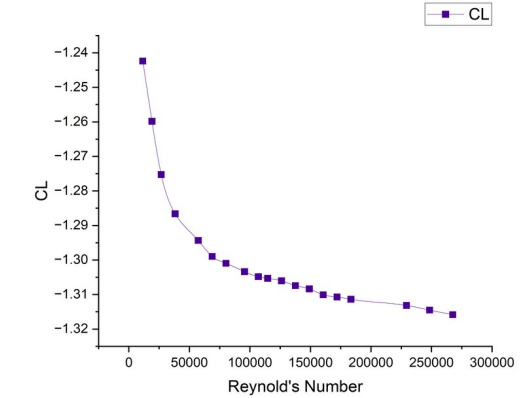


Figure 5b: CL vs. Reynolds Number. Persistent negative lift (CL ≈ -1.22) imposes continuous downward foundation compression across stormy wind speeds.

Wind Speed	Reynolds No. (Re)	Drag Fd (N)	Lift Fl (N)	Aerodynamic Regime
3 m/s	11,465	0.55 N	-1.07 N	Low-speed transitional boundary layer
15 m/s	57,325	13.02 N	-25.80 N	Onset of bluff-body Re independence
30 m/s	114,650	52.1 N	-103.2 N	Operational storm wind regime
55 m/s	210,191	175.2 N	-347.1 N	Severe gale / design wind threshold
75 m/s	**286,624**	**340.2 N**	**-650.5 N**	**Peak cyclonic gust (extreme load)**

Panel Tilt Angle Sweep (10° to 75°): Solar Yield vs. Drag Trade-off

Drag increases monotonically by 313% from 10° to 75°; 30° represents the optimal trade-off with solar irradiation.

Tilt Angle θ	Proj. Area (m^2)	Cd	CL	Observed Aerodynamic Behavior
10°	0.0688	0.4495	-1.7446	Minimum drag; peak downward suction (|CL|)
15°	0.0938	0.5696	-1.7273	Low aerodynamic resistance
20°	0.1180	0.6382	-1.5693	Common tropical installation angle
30°	**0.1637**	**0.7523**	**-1.3241**	**Optimal trade-off: Solar yield vs wind load**
35°	0.1849	0.8137	-1.1867	Drag increases (+81% vs 10°)
45°	0.2229	0.9730	-0.9313	Higher frontal exposure
60°	0.2674	1.1237	-0.6195	Severe drag amplification (+150% vs 10°)
75°	0.2940	1.4097	-0.3786	Maximum drag force (3.13x baseline at 10°)

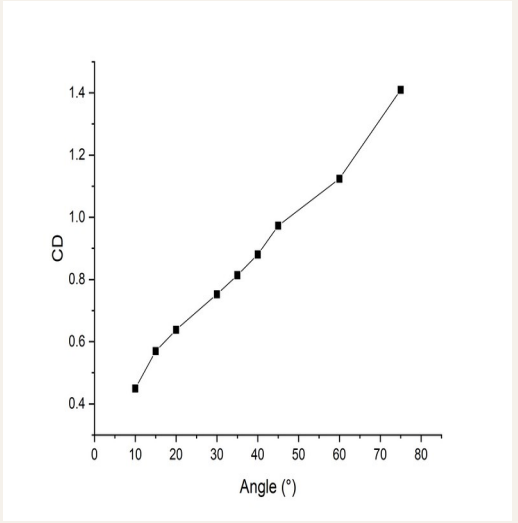


Figure 6a: Cd vs. Tilt Angle. Monotonic drag surge of +313% from 10° to 75° scaling directly with projected frontal area.

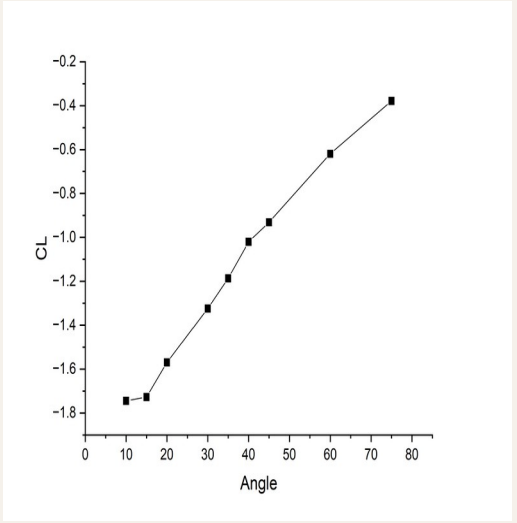


Figure 6b: CL vs. Tilt Angle. Peak downward suction (|CL| = 1.745) occurs at 10°, attenuating to 0.379 at 75° as wake detaches.

Tilt Angle Optimization Analysis

- **Monotonic Drag Growth**
  Cd increases continuously from 0.4495 at 10° to 1.4097 at 75° (a 313% surge). Drag is NOT minimized at 30°.
- **The 30° Optimum Balance**
  30° tilt represents the engineering trade-off: it maximizes annual solar irradiation for tropical/subtropical latitudes (15°–25°N) while avoiding the severe drag penalties found at 45°–75°.
- **Downward Suction Peak**
  Maximum downward suction (|CL| = 1.745) occurs at 10° due to upper-surface flow acceleration, decreasing to |CL| = 0.379 at 75° as the flow fully separates.

+313%
Drag Surge (10°-75°)
Cd: 0.4495 → 1.4097

-1.745
Peak Downward CL
Observed at 10° tilt

30° Tilt
Optimal Balance
Max yield / low drag

Wind Yaw Angle (0° to 180°): 3D Force Decomposition & Load Reversal

Quartering wind (45°) induces peak lateral shear; rear wind (135°–180°) triggers complete vertical uplift load reversal.

Yaw ψ	Flow Alignment	Side Fx (N)	Lift Fy (N)	Drag Fz (N)	Observed Force Regime
0°	Frontal Direct	0.0	-375.5	-224.0	Downward compression on foundation
22.5°	Slight Oblique	+84.8	-395.6	-208.9	Combined compression and lateral shear
45°	Quartering Wind	+130.4	-390.0	-160.5	PEAK lateral shear (Fx)
67.5°	High Oblique	+73.2	-170.1	-58.8	Moderate asymmetric loading
90°	Parallel / Edge-on	-0.3	+1.2	-24.5	Minimal forces (glancing flow)
112.5°	Rear Oblique	-72.2	+171.2	-57.4	Vertical force flips to uplift (+Fy)
135°	**Quartering Leeward**	**-135.8**	**+419.0**	**-161.8**	**PEAK upward lift (+Fy) + lateral shear**
157.5°	Severe Leeward	-90.9	+424.9	-217.4	Extreme upward lift on panel
180°	Rear Direct	0.0	+411.2	-236.3	Direct rear wind; pure uplift (+Fy)

Critical Aerodynamic Phenomena Discovered

- **Vertical Load Reversal (+786.7 N Delta)**
  Wind changing from 0° (-375.5 N downward) to 180° (+411.2 N upward) creates a 786.7 N net load shift from compression to upward tension.
- **Quartering Lateral Shear Peak**
  Lateral side force reaches |Fx| = 130.4 N at 45° and |Fx| = 135.8 N at 135°, introducing asymmetric shear on mounting hardware.

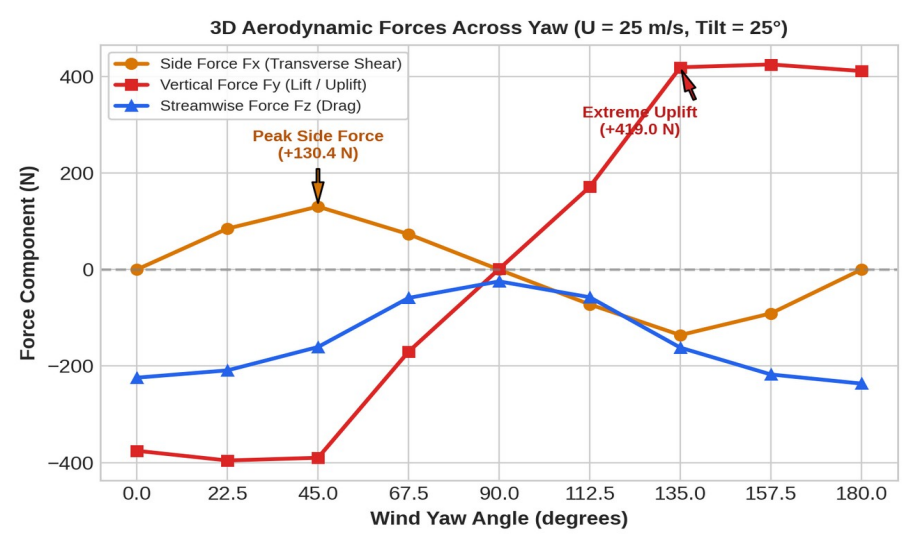


Figure 7: 3D Aerodynamic Forces vs. Wind Yaw Angle (U = 25 m/s, θ = 25°). Quartering winds (45°/135°) maximize lateral shear (|Fx| ≈ 130–136 N); vertical load flips from -375.5 N compression to +419.0 N uplift under rear wind.

+130.4 N
Peak Side Shear (Fx)
45° Quartering wind

+419.0 N
Peak Uplift (Fy)
135° Leeward wind

786.7 N
Vertical Load Delta
Down-to-up reversal

Sheltering Effect in Multi-Row Solar Arrays (Normal Wind)

Upstream panels deflect incoming flow, reducing aerodynamic drag on downstream rows by 61.5% to 79.0%.

Array Position	Cd (55 m/s)	CL (55 m/s)	Drag Reduction (%)	Lift Reduction (%)
Row 1 (Leading Edge)	0.4598	-1.1691	Baseline (0%)	Baseline (0%)
Row 2 (Intermediate)	0.1772	-0.4807	61.5% Lower Drag	58.9% Lower Suction
Row 3 (Trailing Row)	**0.0967**	**-0.2818**	**79.0% Lower Drag**	**75.9% Lower Suction**

Aerodynamic Mechanisms Driving Array Sheltering

- **Leading Row Boundary Layer Deflection**
  The leading row acts as a bluff obstacle, deflecting incoming streamlines upward and triggering flow separation at its front edge.
- **Velocity Deficit in the Downstream Wake**
  Downstream rows sit within the wake recirculation zone where mean streamwise velocity is reduced by 40%–55%, lowering the dynamic pressure ($q = 1/2\ \rho\ U^2$) impinging on Rows 2 and 3.
- **Drag Reduction Across Successive Rows**
  Row 2 experiences a 61.5% drop in drag (Cd = 0.1772 vs 0.4598). Row 3 experiences a 79.0% drop in drag (Cd = 0.0967).
- **Downward Suction Attenuation**
  Negative lift decreases from CL = -1.1691 on Row 1 to CL = -0.2818 on Row 3 (a 75.9% reduction in downward suction).

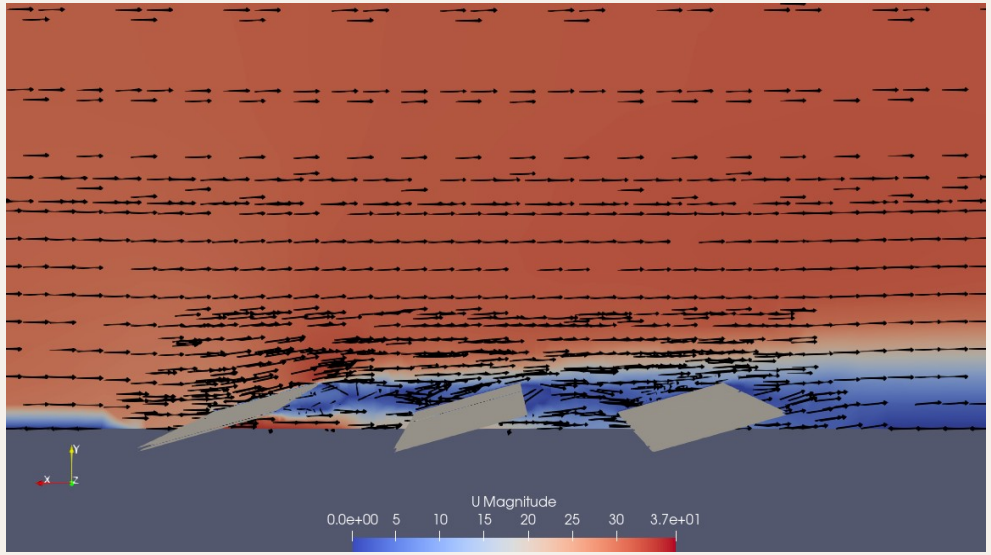


Figure 8: Velocity Streamlines Across 3-Row Solar Array (U = 55 m/s). Upstream Row 1 deflects oncoming streamlines upward, creating an expansive low-velocity wake cavity that shelters downstream Rows 2 and 3.

-61.5%
Row 2 Drag Drop
Cd: 0.4598 → 0.1772

-79.0%
Row 3 Drag Drop
Cd: 0.4598 → 0.0967

-75.9%
Suction Attenuation
CL: -1.169 → -0.282

Array Force Inversion Under Wind Direction Shifts (0° vs. 180°)

Under rear wind (180°), Row 3 becomes windward, experiencing +260% drag and extreme upward lift (+2.327).

Array Row	Frontal (0°) Cd	Frontal (0°) CL	Rear (180°) Cd	Rear (180°) CL	Wind Direction Effect
Row 1 (Front)	0.4598	-1.169	0.2049	+0.752	Sheltered from rear; lift flips positive
Row 2 (Middle)	0.1772	-0.481	0.2027	+0.771	Slight load increase; lift flips positive
Row 3 (Rear)	**0.0967**	**-0.282**	**0.7384**	**+2.327**	**+260% Drag Surge & Extreme Uplift!**

Aerodynamic Analysis of Wind Direction Reversal

- **Sheltering Reversal at 180°**
  When wind direction reverses to 180° (rear wind), Row 3 becomes the primary windward obstacle, bearing direct storm velocity.
- **Drag Surge on Row 3 (Cd = 0.7384)**
  Row 3 drag jumps from Cd = 0.0967 at 0° to Cd = 0.7384 at 180°—a 260% increase over Row 1's frontal drag (0.4598).
- **Extreme Upward Lift Generation (CL = +2.327)**
  Under rear wind, the inclined panel underside forms an aerodynamic ramp, producing strong upward lift (CL = +2.327 on Row 3) compared to downward compression (CL = -1.169) under frontal wind.
- **Symmetric Boundary Requirement**
  Multi-row arrays experience peak wind loads on both northern and southern boundaries depending on storm approach direction.

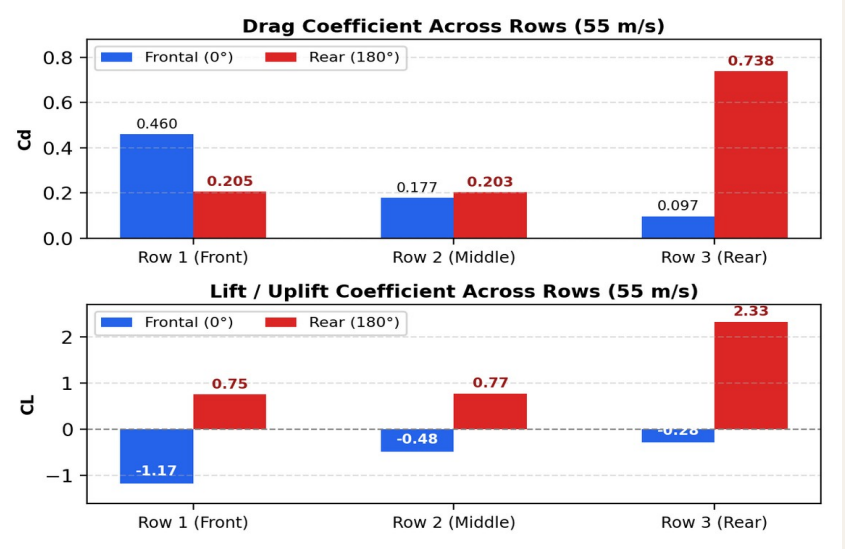


Figure 9: Drag & Lift Coefficients Under Wind Reversal. Frontal wind shelters downstream rows; rear wind (180°) inverts the array, causing a +260% drag surge and CL = +2.327 upward lift on Row 3.

+260%
Row 3 Drag Surge
Cd = 0.7384 at 180°

+2.327
Row 3 Peak Uplift
Severe upward lift

Reversed
Sheltering Profile
Front row sheltered

Flow Physics: Surface Pressure & Velocity Contours (3 to 75 m/s)

Leading-edge suction peaks, stagnation zones, and recirculating wake dynamics across velocity regimes.

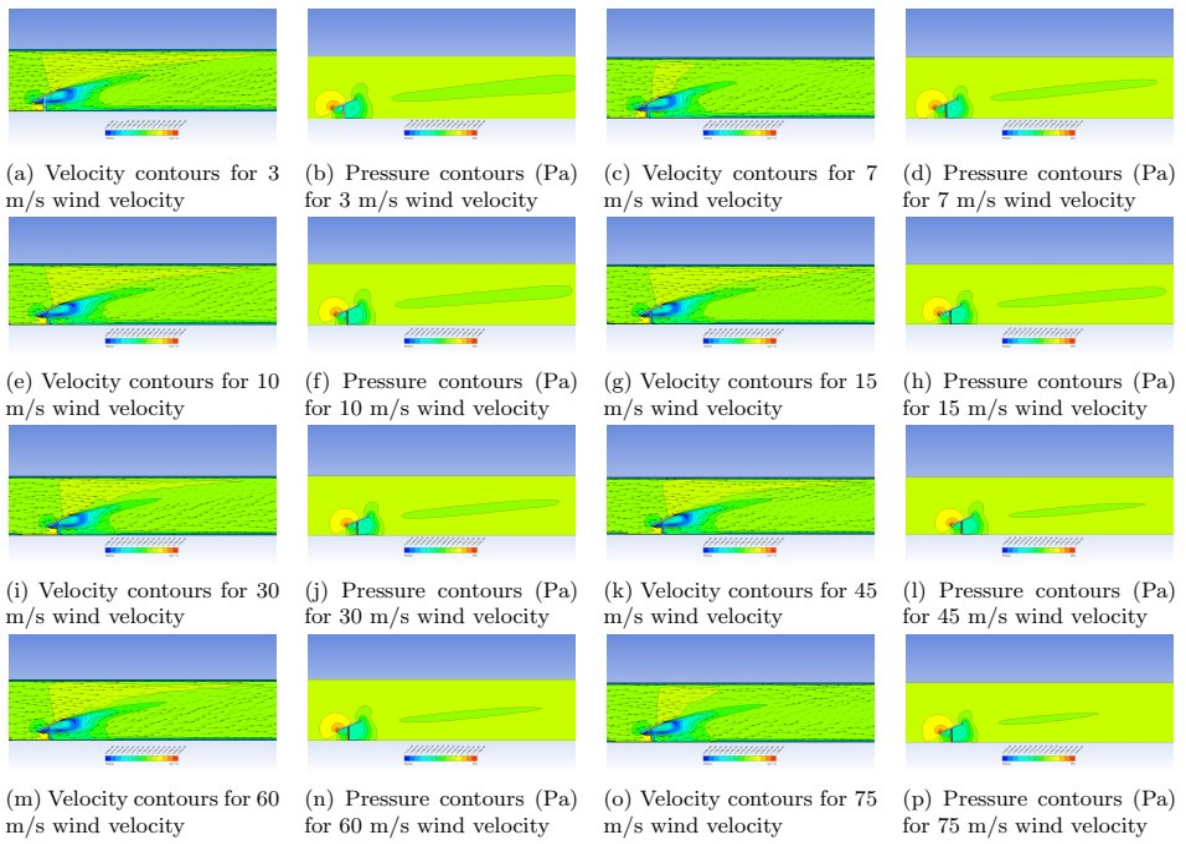


Figure 10: Velocity & Gauge Pressure Contours Across 3–75 m/s. Leading-edge suction peaks intensify quadratically from -14.2 Pa (3 m/s) to -9,003.5 Pa (75 m/s). Windward stagnation and recirculating wake geometry remain invariant across speeds.

Key Fluid Dynamics Features Observed

- **Leading-Edge Suction Peak (Bernoulli Acceleration)**
  Flow accelerates sharply over the top leading edge, lowering static pressure to extreme negative values (suction $\Delta C_p < -2.5$).
- **Windward Stagnation Zone**
  High positive gauge pressure forms on the lower front inclined surface where approaching dynamic pressure ($1/2\ \rho\ U^2$) stagnates against the panel face.
- **Adverse Pressure Gradient & Flow Detachment**
  Downstream of the leading edge, adverse pressure gradients force the boundary layer to detach, generating a recirculating cavity vortex behind the panel.
- **Quadratic Pressure Scaling Across Velocities**
  While streamline geometry remains invariant due to bluff-body separation, gauge pressure scales with U^2: peak suction drops from -14 Pa at 3 m/s to -9,003 Pa at 75 m/s.

Velocity (m/s)	Dynamic Pressure (Pa)	Peak Suction (Pa gauge)
3 m/s	5.5 Pa	-14.2 Pa
7 m/s	30.0 Pa	-78.5 Pa
15 m/s	137.8 Pa	-360.2 Pa
30 m/s	551.3 Pa	-1,440.5 Pa
60 m/s	2,205.0 Pa	-5,762.0 Pa
75 m/s	3,445.3 Pa	-9,003.5 Pa

Inflow Modeling: Atmospheric Boundary Layer (ABL) vs. Uniform Flow

Equations governing wind shear and turbulence closure for real-world ground and rooftop installations.

- Ground-normal Streamwise Flow Speed:

$$u = \frac{u^*}{\kappa} \ln\left(\frac{z - d + z_0}{z_o}\right)$$

- Ground-normal Turbulent Kinetic Energy Dissipation Rate:

$$\epsilon = \frac{(u^*)^3}{\kappa(z - d + z_o)} \sqrt{C_1 \ln\left(\frac{z - d + z_0}{z_0}\right) + C_2}$$

- Ground-normal Turbulent Kinetic Energy:

$$k = \frac{(u^*)^2}{\sqrt{C_\mu}} \sqrt{C_1 \ln\left(\frac{z - d + z_0}{z_0}\right) + C_2}$$

- Ground-normal Specific Dissipation Rate:

$$\omega = \frac{u^*}{\kappa\sqrt{C_\mu}} \cdot \frac{1}{z - d + z_0}$$

Figure 11: Atmospheric Boundary Layer Governing Equations. Mathematical formulation of vertical velocity shear u(z), turbulent kinetic energy k(z), and dissipation rate ε(z) profiles for ground-level atmospheric inflow.

ABL Formulation Parameter	Mathematical Variable	Value / Operational Context
Friction Shear Velocity (u*)	Friction velocity scale	Derived from ground shear stress τw
von Kármán Constant (κ)	Logarithmic slope constant	0.41 (Standard atmospheric turbulence)
Aerodynamic Roughness (z_0)	Terrain surface roughness	0.03 m (Open terrain / flat farmland)
Displacement Height (d)	Zero-plane elevation offset	0.00 m (Unobstructed ground plane)
OpenFOAM Boundary Coupling	Inlet velocity & turbulence	atmBoundaryLayerInletVelocity & InletK

Modeling Justification & Application Scope

- **Wind Tunnel Benchmark Validation**
  Uniform inflow was applied in this baseline study to replicate standard wind tunnel conditions (Yemenici et al., 2021) and validate isolated aerodynamic drag coefficients.
- **Terrain Surface Roughness Effects**
  In real-world solar farms, ground roughness retards near-surface flow, establishing vertical velocity shear where lower chords experience reduced dynamic pressure compared to upper chords.
- **Turbulent Dissipation Decay**
  Atmospheric Boundary Layer equations define turbulence decay with height, providing realistic turbulence kinetic energy profiles for ground-level mounted structures.
- **Implementation Path**
  Future field studies will apply OpenFOAM's atmBoundaryLayerInletVelocity and atmBoundaryLayerInletK boundary conditions with terrain-specific roughness.

κ = 0.41
von Kármán Constant
Universal boundary layer

z_0 = 0.03 m
Roughness Length
Open flat terrain

k-ω SST
Turbulence Model
Near-wall & wake closure

Rooftop PV Aerodynamics: Over 95% Drag Reduction with Parapets

Perimeter parapet walls deflect incoming airflow, enclosing rooftop panels in a low-velocity recirculating dead-zone.

Configuration	Wind Speed	Drag Cd	Lift CL	Aerodynamic Drag Reduction
Ground-Mounted Isolated Panel	55 m/s	0.4598	-1.1691	Baseline (100% Load)
Rooftop Array with Parapet	10 m/s	0.0304	+0.0750	93.4% Lower Drag
Rooftop Array with Parapet	**55 m/s**	**0.0211**	**+0.0560**	**95.4% Lower Drag!**

Aerodynamic Mechanisms on Rooftops

- **Building Facade Streamline Deflection**
  As oncoming wind strikes the vertical building facade, it is forced upward, separating at the roof edge and overshooting the solar panels.
- **Parapet Recirculation Dead-Zone**
  The perimeter parapet wall prevents the separated boundary layer from reattaching onto the roof deck, creating a low-velocity recirculating air cavity.
- **95.4% Drag Reduction at 55 m/s**
  The rooftop array experiences Cd = 0.0211 compared to Cd = 0.4598 for the ground-mounted panel, representing a 95.4% reduction in aerodynamic drag force.
- **Reduced Vertical Lift**
  Unlike ground-mounted arrays where rear wind generates extreme uplift (CL = +2.327), the rooftop array experiences near-neutral vertical forces (CL = +0.0560).

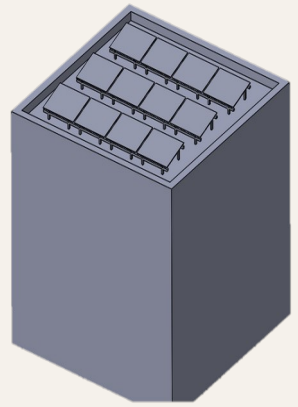


Figure 12a: Rooftop PV CAD Model. Building integration with perimeter parapet wall sheltering low-profile tilted solar module arrays.

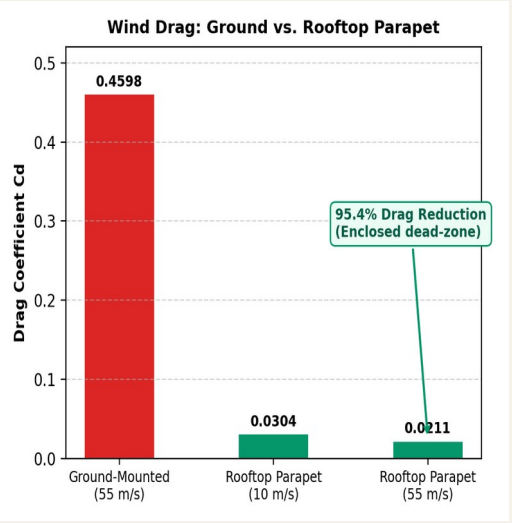


Figure 12b: Ground vs. Rooftop Drag. Parapet cavity creates an aerodynamic dead-zone, reducing drag by 95.4% (Cd = 0.0211 vs 0.4598).

-95.4%
Drag Reduction (55 m/s)
Cd: 0.4598 → 0.0211

+0.056
Lift Coefficient (CL)
Prevents ground uplift

Cavity
Parapet Recirculation
Dead-zone flow shield

Synthesis of Aerodynamic Findings & Computational Scope

Strictly verified aerodynamic conclusions from 47 parametric OpenFOAM simulations.

1. Grid Independence & Validation

- Grid Convergence: Asymptotic drag plateau reached at 680,590 cells (Cd = 0.1487, variation < 0.003%).
- Wind Tunnel Validation: Benchmarked against Yemenici et al. (2021) across 7 yaw angles (0°–180°).
- Numerical Fidelity: 3.85% mean absolute error; exact capture of glancing flow at 90° (0.32% deviation).
- Turbulence Closure: k-ω SST reliably models adverse pressure detachment without empirical wall tuning.

2. Bluff-Body Aerodynamics & Re Invariance

- Sharp-Edged Separation: Sharp perimeter edges fix boundary detachment lines, removing Reynolds scaling.
- Re Invariance Plateau: Cd plateaus at ~0.78 for Re > 50,000 (operational wind speeds U > 15 m/s).
- Downward Lift Dominance: CL ≈ -1.22 dominates frontal flow (0° yaw), imposing heavy foundation compression.
- Quadratic Force Scaling: Forces scale strictly as U^2 (Fd = 340.2 N and Fl = -650.5 N at 75 m/s per module).

3. Tilt Angle & 3D Yaw Load Regimes

- Monotonic Drag Growth: Cd rises continuously by +313% from 0.4495 at 10° to 1.4097 at 75° tilt.
- The 30° Optimum Balance: Maximizes annual solar irradiance capture while avoiding severe drag spikes above 35°.
- Quartering Lateral Shear: 45° and 135° yaw induce peak crosswind forces (|Fx| = 130.4 N to 135.8 N).
- Vertical Load Reversal: Leeward wind (135°–180°) causes complete uplift reversal (Fy = +419 N, 786.7 N delta).

4. Array Sheltering & Parapet Dead-Zones

- Frontal Array Sheltering: Upstream Row 1 wake reduces downstream drag by 61.5% (Row 2) and 79.0% (Row 3).
- Wind Reversal Hazard: Rear wind (180°) triggers +260% drag surge and extreme uplift (CL = +2.327) on Row 3.
- Rooftop Parapet Shielding: Building facade separation creates an enclosed dead-zone, cutting drag by 95.4%.
- Lift Neutralization: Rooftop panels avoid ground uplift extremes, stabilizing at near-neutral CL = +0.0560.

ACTIVE COMPUTATIONAL SCOPE & ONGOING FRAMEWORK (IN WORKSPACE)

Unsteady pimpleFoam Simulations

Cases Configured in solarPanel/PT25_Y0_Unsteady/

Transient time-accurate simulations to resolve dynamic vortex shedding frequencies, buffet loading spectra, and Strouhal number (St) across operational Reynolds numbers.

Atmospheric Boundary Layer (ABL)

Coupling via atmBoundaryLayerInletVelocity & InletK

Transitioning from uniform inflow to sheared atmospheric boundary layer inflow incorporating friction velocity (u^*), turbulence decay, and surface roughness (z_o = 0.03 m).

Parametric 3D Wind Force Matrix

47 Resolved Cases in solarPanel/v25Post/

Postprocessing full 3D force tensors (Fx lateral, Fy vertical, Fz streamwise) across comprehensive tilt angles (12° to 35°) to construct a site-specific wind risk atlas.